

Multiple Choice

1. **Answer:** C

Options: A) 1.0/2; B) 1.0/4; C) 1.0/8; D) 1.0/16

Note: The correct answer is 1.0/8 because first cousins share, on average, 12.5% of their genes, which is equivalent to 1 out of every 8 alleles inherited from a common ancestor.

2. **Answer:** B

Options: A) Cab drivers; B) Fire fighters; C) Insurance regulators; D) Computer programmers

Note: Firefighters often face mandatory retirement ages due to the physically demanding nature of their job and the need to ensure that they can perform their duties effectively and safely.

3. **Answer:** B

Options: A) Genetic factors; B) Lifestyle factors like not exercising; C) Hormonal factors such as loss of estrogen; D) Cohort factors

Note: Loss of bone density is most strongly associated with lifestyle factors such as lack of exercise, as physical activity is crucial for stimulating bone formation and maintaining bone strength.

4. **Answer:** D

Options: A) war; B) natural disasters; C) hunger; D) all of these

Note: Intergenerational transmission to offspring can occur as a result of parental exposures to biological, environmental, and behavioral factors, indicating that all these elements collectively influence the health and development of subsequent generations.

5. **Answer:** A

Options: A) Compulsory; B) Unlinked anonymous; C) Voluntary anonymous; D) Voluntary confidential

Note: Compulsory surveillance testing minimizes selection bias by ensuring that all individuals within a defined population are tested, regardless of their health status or willingness, thereby providing a more representative sample of the population.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Needle aspiration is a diagnostic procedure that involves using a thin needle to extract fluid or tissue from a breast lump, and it does not involve exposure to radiation, unlike imaging techniques such as mammography.

7. **Answer:** False

Options: A) True; B) False

Note: A cross-sectional design only captures data at one point in time, making it ineffective for simultaneously analyzing age and cohort differences, as it cannot account for changes over time or the influence of historical events on different cohorts.

8. **Answer:** True

Options: A) True; B) False

Note: Research indicates that age-related decline in olfactory function is common, with studies showing that more than 75% of adults aged 80 and older experience significant loss of smell due to neurological and physiological changes associated with aging.

9. **Answer:** True

Options: A) True; B) False

Note: Bile, produced by the liver and stored in the gallbladder, is released into the duodenum to emulsify fats, facilitating their digestion and absorption in the small intestine.

10. **Answer:** False

Options: A) True; B) False

Note: A pulse cannot be felt on a patient's face at that location because the superficial temporal artery typically lies deeper and is not directly palpable at the point where it crosses the maxilla.

Long Form Response

11. **Answer:** Surprise visits to grandparents can significantly enhance the emotional well-being of college students by providing unexpected moments of joy and connection. These visits often foster a sense of belonging and support, which is crucial during the stressful college years. Additionally, seeking advice from grandparents allows students to tap into their rich life experiences and wisdom, promoting resilience and coping strategies in challenging situations. This intergenerational interaction not only strengthens familial bonds but also fosters mutual understanding and respect, enriching the emotional landscape for both students and their grandparents. Ultimately, these practices cultivate deeper relationships and contribute positively to the mental health of college students.

Note: The answer correctly highlights that surprise visits create joy and support for students, while seeking advice from grandparents leverages their life experience to enhance resilience, thereby strengthening intergenerational relationships and improving emotional well-being.

12. **Answer:** Azidothymidine (AZT) inhibits herpes viruses primarily through its mechanism as a nucleoside analog, which interferes with viral DNA synthesis. By mimicking thymidine, AZT gets incorporated into the viral DNA, leading to premature chain termination during replication. While AZT has shown some effectiveness against herpes viruses, especially in immunocompromised patients, its use is often limited due to

side effects and the emergence of resistance. Compared to other antiviral treatments, such as acyclovir, which specifically targets the herpes virus enzyme thymidine kinase, AZT is generally less effective and not the first-line treatment for managing herpes infections. Overall, while AZT can provide some antiviral activity, its role in herpes management is overshadowed by more targeted therapies that offer better efficacy and safety profiles.

Note: correct because it accurately describes how azidothymidine functions as a nucleoside analog to inhibit viral DNA synthesis in herpes viruses, while also acknowledging its limitations in effectiveness and the potential for resistance compared to other antiviral treatments.

End of Answer Key